ARYAMAN BHATNAGAR

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Portfolio: https://aryamanbhatnagarcad-portfolio.github.io/Aryaman-Bhatnagar/

PERSONAL STATEMENT

Stress engineer focusing on composite structures, finite element analysis, and weight reduction strategies, with an MSc in Automotive Engineering from Cranfield University. Skilled in linear, nonlinear, and explicit dynamic analyses through HyperMesh, OptiStruct, ANSYS, and LS-DYNA. Extensive knowledge of CFRP structures, laminate design, and failure predictions using Tsai-Wu criteria. Familiar with fatigue, modal, and crash simulations in performance-oriented engineering settings. Looking for opportunities to contribute to structural safety in the aerospace, defense, and motorsport industries.

TECHNICAL SKILLS

Stress & CAE

- Linear & Nonlinear FEA, Explicit Dynamics (impact/crash)
- Fatigue, Buckling, Modal & Transient Analysis
- Load case definition, correlation, and validation
- Hand calculations (beam theory, stress equations)
- Composites
- CFRP laminate modelling, stacking sequence optimisation
- Sandwich structures (honeycomb cores)
- Failure criteria: Tsai-Wu, Hashin (familiar), von Mises
- Damage tolerance and stiffness-to-weight optimisation

Tools

- ABAQUS (Advanced)
- Altair HyperMesh / OptiStruct (Advanced)
- ANSYS Mechanical (Advanced)
- LS-DYNA, HyperCrash (Intermediate)

KEY ACHIEVEMENTS

- 15% UAV mass reduction using topology optimisation, maintaining a FOS of 3.5 (AERO2ASTRO, 2021)
- 4.89% reduction in drag of a rider-motorcycle assembly through strategic placement of passive flow devices using Advanced CFD (Cranfield University, 2025)
- 25% weight reduction while maintaining a FOS of 3.2 (Cranfield University and McLaren, 2025)
- Conducted CAD and Engineering Drawing Workshops for 65+ students (Manipal University Jaipur, 2021)

ACADEMIC PROJECTS

Vehicle Structures Module in collaboration with McLaren Automotive **January 2025 - February 2025**

- Redesigned the McLaren W1 subframe and performed linear FEA using HyperMesh Optistruct.
- Performed stress equation hand calculations to validate simulation results with ~10% deviation
- Achieved 25% weight reduction with FOS 3.2, ensuring structural safety and performance
- Conducted fatigue and modal analysis under cyclic loading
- Performed non-linear and transient dynamic analysis using HyperCrash, achieving ~200 MPa peak von Mises stress while maintaining FOS ~1.9
- Conducted fatigue analysis and applied Fatigue & Damage Tolerance (F&DT) principles under cyclic loading conditions

CAREER HISTORY

Race Anywhere, Daventry, United Kingdom- Design Engineer **November 2025 - December 2025**
The role ended due to management restructuring and the dissolution of the design team

- Performed linear static FEA on mounts, evaluating stress distribution and von Mises stress regions ~120 MPa under operational loads
- Reduced 35% mass of handbrake mount through topology optimisation with a FOS of 4 validated via ANSYS
- Managed 8% correlation between hand calculations using beam theory and stress equations and FEA results
- Produced detailed technical reports and documentation to communicate FEA results, assumptions, and design recommendations

- Designed production-ready mounts and brackets and applied DFM principles for CNC machining using SolidWorks

Schindler Group, Mumbai, India- Executive- Quality Assurance **July 2022 - May 2024**

- Reduced callbacks of the S3000 by 7% by conducting root cause analysis of mechanical components
- Improved the carbon emissions of maintenance vehicles by 3.84% by deploying 50+ electric vehicles
- Worked within ISO-compliant QMS, EMS, and H&S standards, supporting engineering processes aligned with regulated industry standards
- Reduced ISO Audit non-conformities by 40% by conducting a stringent internal audit and root cause analysis

AERO2ASTRO, Tamil Nadu, India- Design Engineering Intern **November 2020 - April 2021**

- Achieved 7× improvement in stiffness-to-weight ratio by developing a CFRP honeycomb sandwich structure
- Applied composite laminate theory and Tsai-Wu failure criteria to validate structural integrity under aerodynamic and payload loads
- Managed a FOS of 3.5 by performing stress analysis and load case evaluation under operational conditions
- Supported DFM of composite components, considering layup feasibility and material behaviour

Infi League Motorsport, New Delhi, India- Chassis Design Engineering Intern **August 2020 - September 2020**

- Designed composite monocoque chassis in CATIA V5, defining laminate stacking sequences and ply orientations
- Conducted ply-level failure analysis using Tsai-Wu criteria to ensure structural safety
- Achieved 18 kg mass reduction while maintaining torsional stiffness targets
- Integrated FEA-driven optimisation into the structural design process

EDUCATION

MSc in Automotive Engineering, Cranfield University, Cranfield, UK **October 2024 - September 2025**

- **Key Modules:** Vehicle Design, Vehicle Structures, Composites, FEA, Material and Manufacturing

B.Tech Mechanical Engineering, Manipal University, Jaipur, Rajasthan, India **June 2018 - July 2022**

- **Key Modules:** Solid Mechanics, Thermodynamics, Fluid Mechanics, Machine Design, and Material Science

ADDITIONAL INFORMATION

- **Languages:** English (Fluent), Hindi (Native)
- **Right to Work:** Graduate visa (No sponsorship required till December 2027)